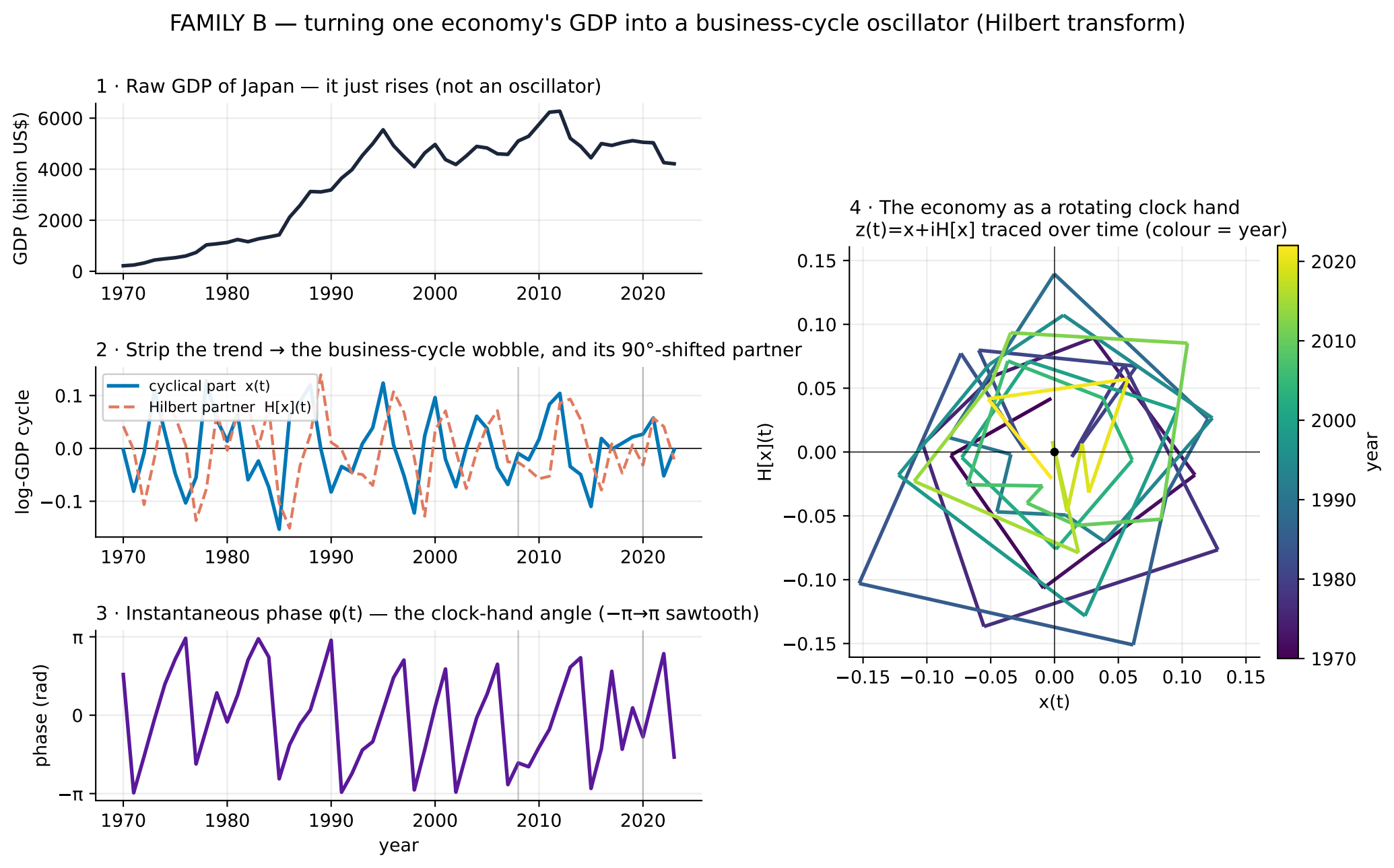
FAMILY B — turning one economy's GDP into a business-cycle oscillator (Hilbert transform)
1 · Raw GDP of Japan — it just rises (not an oscillator)
GDP (billion US$)
0
2000
4000
6000
1970
1980
1990
2000
2010
2020
2 · Strip the trend → the business-cycle wobble, and its 90°-shifted partner
log-GDP cycle
cyclical part x(t)
Hilbert partner H[x](t)
0.1
0.0
−0.1
3 · Instantaneous phase φ(t) — the clock-hand angle (−π→π sawtooth)
phase (rad)
π
0
−π
year
4 · The economy as a rotating clock hand
z(t)=x+iH[x] traced over time (colour = year)
H[x](t)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
x(t)
year
2020
2010
2000
1990
1980
1970